

AI/ML Engineer Assignment

Problem Statement

Organizations often struggle to find information quickly when knowledge is distributed across multiple documents such as reports, manuals, policies, FAQs, and technical documentation.

Your task is to build an intelligent system that enables users to upload a collection of documents and interact with them through natural language questions.

The system should be capable of understanding user queries, locating relevant information from the uploaded documents, and generating accurate, concise, and context-aware responses.

Requirements

Document Management

Users should be able to:

Upload one or more documents.

View uploaded documents.

Supported file formats: PDF, TXT, DOCX

Intelligent Question Answering

Users should be able to ask questions in natural language related to the uploaded documents.

Examples:

"What are the responsibilities of a data science intern?"

The system should return:

A clear answer to the user's question.

The system should:

Handle multiple documents efficiently.

Produce responses within a reasonable time.

Remain functional as the number of documents grows.

Deliverables:

1. Source Code (GitHub Repository)

2. README containing:

Setup Instructions

Architecture Overview

Assumptions Made

3. Sample Documents for Testing

4. Docker Setup (Optional)